Linuxzoo Tutorial Answers

Intro 1

Q2. cal
Use the cal command to find out which day the 31st of December 2002 was. Cal takes two parameters, the number of the month (e.g. 1 for January or 8 for August) and the year as a four-digit number (e.g. 1997).
$ cal 12 2002

Q3. cal year
Cal can also work with one parameter, which is the year. When given a year, it displays a calendar for the whole year. Use this feature but redirect the output of cal so that the calendar for 2005 is saved to a file called “yearfile” in the home directory of demo.
$ cal 2005 > /home/demo/yearfile

Q4. ls
Use the ls command to see the hidden and normal files you have in the home directory for “demo”. Count the files and directories shown. Include the “.” and “..” files.
$ ls -ltra

Q5. file size
Which file in /home/demo (including hidden files) has the smallest size.
$ ls -ltra

Q6. append
Use the redirection to create a file called “this month”, containing only this month’s current calendar. Then use the date command to append the current date to the end of the same file.
$ cal 9 2019 > thismonth
$ date >> thismonth

Q7. copying
Copy the file yearfile to yearfile2. Copy the file yearfile to yearfile3.
$ cp yearfile yearfile2
$ cp yearfile yearfile3

Q8. moving
Change the name of file yearfile3 to thisyear.
$ mv yearfile3 thisyear

Q9. deleting
Delete the file yearfile.
$ rm yearfile

Q10. big concat
Concatenate the files “thismonth”, “yearfile2” and “thisyear” together to form a single file called “bigfile”. Check your command worked using more.
$ cat thismonth yearfile2 thisyear > bigfile
$ more bigfile

Intro 2

Q2. Create a directory structure
In the demo account /home/demo create the directory structure shown:

The top directory of the structure (demo) corresponds to /home/demo, your HOME directory and therefore the directory you start in when you log into your machine using the demo username.
You do not create a “demo” directory yourself to answer this question, as it already exists are you are already in it.
$ mkdir work letters scripts
$ cd work
$ mkdir progs tutorial misc

Q3. cp
Copy the files /etc/group and /etc/vimrc into your misc directory.
$ cp ./etc/group ./work/misc
$ cp ./etc/vimrc ./work/misc

Q4. Relative move

- Make misc your current directory
- Move the file vimrc to the progs directory
- Use ls to verify that this has worked

$ cd work/misc
$ mv vimrc ../progs

Q5. rename
While still in the misc directory using a relative path name, copy the file “bigfile” from your home directory to the tutorial directory and rename it “bigfile2”.
$ cp ../../bigfile ../tutorial/bigfile2

Q6. cp
Make the directory work your current directory and using an absolute path name, copy the file “bigfile2” in the tutorial directory to the scripts directory.
$ cd ..
$ cp ./tutorial/bigfile2 /home/demo/scripts/

Q7. tilde
We may use ~root for the home directory of the user root. Use cd and pwd to find the home directory of the user mysql.
$ ~mysql

Q8. case and space
We can have spaces in our file and directory names (like Windows). File names and directory names are case sensitive (unlike Windows). Create directories with the following names in your home directory, use ls to check they are there.
$ mkdir ‘My Documents’ gordon Gordon

BasicShell

Question 2: cp, mv and cat

COPY file1 from test1 to data. Keep the name as file1.
$ cp test1/file1 data/

COPY file2 from test1 to data. Change the name as you copy the file to the new name of filecopy1.
$ cp test/file2 data/filecopy1

Rename info1 to newinfo1. Do not move it out of mydir1.
$ mv mydir1/info1 mydir1/newinfo1

Change directory into mydir1, and then copy “lines” into the current directory.
$ cd mydir1
$ cp ../lines .

Still in mydir1, concatenate info2 and lines, saving the output as “joined”.
$ cat info2 lines > joined

Still in mydir1, concatenate info2 and lines and file1 from test1, saving the output as “joined2”.
$ cat info2 lines ../test1/file1 > joined2

Make a new directory in /home/demo called "etccopy". Now copy all of the files which begin with a "u" and end with ".conf" from /etc to your new directory. Only copy from /etc and not from any sub-directories (i.e. not recursively).
$ mkdir etccopy
$ cp /etc/u*.conf etccopy/

List all the file and directory names in /etc which have 1 or more numbers in their name. Do not descend into subdirectories (i.e. no recursion). Remember to use "-d" in the ls command. Save this list in demo's home directory as a file called "digit". Do the "ls" command when CDed to /etc, thus your names avoid having "/etc/" stuck in front of them.
$ cd /etc
$ ls -d *[0-9]*

Redo the previous question, except this time repeat the exercise when CDed in /home/demo. Save the output in digit2. "digit" and "digit2" should be identical, except in "digit2" all the filenames will begin "/etc/".
$cd /home/demo
$ ls –d /etc/*[0-9]* > /home/demo/digit2

Question 3: Head and Tail

In this question you will be practicing with the head and tail command. Remember these commands take a parameter (starting with a minus) which indicates how many lines from the head or tail to print to the screen.

As an example, consider:

head -1 /etc/passwd

The above command will give one line from the head (start) of the file /etc/password.

What is the first line printed if you tail the last 10 lines of the file "lines"... i.e. what is the line 10 lines from the end of "lines"?
$ tail -10 lines | head -1

What is the word which appears on line 5000 of the file "words". Use a pipe with head and tail to make life easy... The test is CASE SENSITIVE
$ head -5000 words | tail -1

If you do a "ls" in /etc, what is the 100th filename?
$ ls /etc | head -100

If you do a "ls" in /etc, what is the 100th filename from the END?
$ ls /etc | tail -100 | head -1

Using head and tail, store the filenames printed on lines 120 to 150 inclusive when doing a ls in "/etc". Save the names in a file called /home/demo/120. Hint: there are 31 lines from 120 to 150 inclusive...
$ ls /etc | head -150 | tail -31 > 120

In the concatination of file1, file2, file3, and file4 in test1, how many characters are there on line 167? Hint: use the wc command to count the characters (thus include the newline character).
$ cat file1 file2 file3 file4 | head -167 | tail -1 | wc -c

Question 4: Permissions and chmod

In this question you are reqired to demonstrate your knowledge of symbolic and numeric permissions, both in reading current settings and in setting permissions for existing files. What are the current symbolic permissions on "info" in /home/demo? Use the format "-rwxrwxrwx". Do not include any extended permissions if any are present
$ ls -l info

What is this permission if it was encoded in numeric (octal) notation? Use the form 0666.
0644

Change the permissions on info so that user demo can read, write, and execute, group tutorial can read and execute, and other can write and execute.
$ chmod u+x info
$ chmod g+x info
$ chmod o-r info
$ chmod o+wx info
$ ls -l info

$ chmod u=rwx,g=rx,o=wx info

$ chmod 753 info

Do the follow question in your head. Do not actually do this to the info file.

Assume you now have to set the file info using numeric notation. Preserving all permissions except group, what would be the numeric notation for giving group just execute permissions? Again, use the format 0777.
0715

Now in Linuxzoo, take away write permissions for other on the info file.
$ chmod o-w info

Change the permissions again of info so that anyone in group tutorial can read the file but not write or execute it. Leave the other permissions unchanged.
$ chmod g-x info

Given a file owned by "gordon" in group "users", and a numeric permission code of 0654, what would user "demo" of group "demo" be allowed to do?
Read Access

Given a file owned by "gordon" in group "demo", and a numeric permission code of 0654, what would user "demo" of group "demo" be allowed to do?
Read and Execute Access

Given a file owned by "demo" in group "users", and a symbolic permission code of "-rw-rwx--x, what would user "demo" of group "demo" be allowed to do?
Read and Write Access

Adjust the info file once more so that a user "gordon" in group "tutorial" would only be allowed to write and execute the file info. Again, do not change any other file permissions other than that required to answer this question. It might be an idea to note the current permissions just in case you have to try again.
$ chmod g-r info
$ chmod g+wx info
$ ls -l

Adjust the permissions on info once again, so that group would have the octal permission 4, and the owner would have the symbolic permission of "rw", and others would have permissions to read only.

```
$ chmod u-x info
$ chmod g-wx info
$ chmod g+r info
$ chmod o-x info
$ chmod o+r info
$ ls –l

$ chmod u=rw,g=r,o=r info
```

Search

Basic File Analysis

Question 2: Signatures

Using the “file” command, evaluate the file signature of theanalysis/file1.
What is file1?
$ file theanalysis/file1
file1 is an ELF 64-bit file. ELF stands for Extensible Linking Format, which is a common standard file format for executable files, object code, shared libraries and core dumps.

Using the “file” command, evaluate the signature of theanalysis/file4.
What is file4?
$ file theanalysis/file4
file4 is data.

Use the md5sum command to calculate the md5 hash of file3.
What is the hash of file3?
$ md5sum theanalysis/file3
Hash = 79054025255fb1a26e4bc422aef54eb4
If another file has the same md5 hash, what does this mean?
Meaning of two files with the same hash is that the file contents are probably identical.

Use the md5sum command to calculate the md5 hash of file4.
What is the md5 hash of file4?
$ md5sum theanalysis/file4
Hash = 79054025255fb1a26e4bc422aef54eb4

Use the cmp command to verify that file3 and file4 are identical.
$ cmp theanalysis/file3 theanalysis/file4
Output of the cmp command = theanalysis/file3 theanalysis/file4 differ: byte 20, line 1

Check file3 and file4 using a 512 bit sha hash.
$ sha512sum theanalysis/file3
$sha512sum theanalysis/file4
What does this mean?
This means that the contents of the files are definitely different.

Question 3: FIND command

Use the find command to locate all files which start with an "a" and end in a ".conf" which exist somewhere in /usr/share. Save this list to a file '/home/demo/alist'. Make sure that the first parameter of find is "/usr/share". Run the command as user "demo" and do not worry about any permission error messages.

$ find /usr/share -name ‘a*.conf’ > /home/demo/alist

Use the find command to locate all FILES in /usr/share which have the permissions "rwxr-xr-x". Save this list to /home/demo/blist. Make sure that the first parameter of find is "/usr/share". Run the command as user "demo" and ignore any permission error messages.

$ find /usr/share -type f -perm 755 > /home/demo/blist

Use the find command to locate all directories in /usr/share which have the permissions "rwxr-xr-x". Pipe this list to wc and count the number of directories.

$ find /usr/share -type d -perm 755 | wc -l

Again, using the find command find out how many FILES in /etc are in group lp.

$ find /etc -type f -group ‘lp’ | wc -l

Question 4: GREP and regexp

Using a combination of grep, regular expressions, and wc via a pipe count how many words in the /usr/share/dict/words dictionary starts with "anti" and ends with an "n".

$ grep -E “^anti.*n$” /usr/share/dict/words | wc -w

Using grep and regular expressions, create a file /home/demo/aword which contains all the words which start with "tele" from /usr/share/dict/words, and which are exactly 7 characters long.

$ grep -E ‘^(tele…)$.*’ /usr/share/dict/words > /home/demo/aword

How many words have the string "ra" in them twice in /usr/share/dict/words?

$ grep -E ‘(ra).*\1’ /usr/share/dict/words | wc -w

How many words are in /usr/share/dict/words which contains "ice" then immediately following that either an "s" or a "d" (i.e. ices or iced). Use square brackets to form a set in your regular expression.

$ grep -E ‘(ice[sd])’ /usr/share/dict/words | wc -w

Use grep on words to find a word that contains each of the vowels in alphabetical (i.e first an A, then an E, etc) order in /usr/share/dict/words. How many such words are there? (you may include words with extra vowels such as adventitious. A vowel is one of A,E,I,O,U.
$ grep -E '*a.*e.*i.*o.*u.*' /usr/share/dict/words | wc -w

The word interlinking includes the same two characters (e.g. "in") which appear three times. The word "priestesses" also contains the same two characters repeated three time (e.g. "es").
How many words can you find which contain any two characters repeated three times, like the examples "interlinking" and "priestesses". Use /usr/share/dict/words as your list of possible words and grep to find the answer.
Note. Do a case sensitive match, so do not use "-i".
$ grep -E '(..)(.*\1){2}' /usr/share/dict/words | wc –w

$ grep -E '(..).*\1.*\1' /usr/share/dict/words | wc –w – taught method.

Challenge Question: This is a tricky question. Just give it 10 minutes before moving onto the next question!
How many words are 5-character palindromes? A palindrome is a word spelled the same way forward and backwards, such as "sagas". Use /usr/share/dict/words. Hint: Use multiple groups and backreferences.
$ grep –E '^(.)(.)./2/1$' /usr/share/dict/words | wc -w

Wildcard

Q2. Wild copy
In the demo account, cd to the home directory (/home/demo). From there copy all files using the * wildcard that contain the word file in the filename into the work directory. Confirm this using ls.
$ cp *file* ./work

Q3. Duplicate thismonth
Copy the file thismonth to the letters directory and rename it let1.doc. Change the current working directory to the letters directory and copy let1.doc to let2.doc. Copy once more let1.doc to let3.doc. Confirm this yourself using the ls command.
$ cp ./thismonth ./letters/let1.doc
$ cd letters
$ cp let1.doc let2.doc
$ cp let1.doc let3.doc
$ ls -ltr

Q4. Copy and rename
Copy using the ? wildcard, the three files let1.doc, let2.doc and let3.doc from the letters directory to the work/misc directory.
Once copied, rename the new files (the ones in the misc directory) into rpt1.doc, rpt2.doc and rpt3.doc. When renaming, you must do this explicitly and not try to use a wildcard. It would appear at first that a wildcard could make the renaming easier, but it turns out to be ridiculously hard to use a wildcard in this case.
$ cp ./let?.doc ../work/misc
$ cd ../work/misc
$ mv let1.doc rpt1.doc
$ mv let2.doc rpt2.doc
$ mv let3.doc rpt3.doc

Q5. Square Brackets
Using square brackets, move all the files from the misc directory that contain the numbers 2 or 3 into the scripts directory.
$ mv *[23]* ../../scripts

Q6. rm
Remove all files beginning with the letter ‘r’ from the scripts directory. Use the -i option to see what you are about to delete before then agreeing to delete the files involved.
$ rm -i ../../scripts/r* - need to type yes before hitting Return

Q7. Hard link
Create a hard link in progs called biglink, which is a hard link to the bigfile in your home directory.
In “ls -l bigfile” the hard link count should be 2. If you make mistakes and create links with the wrong name or in the wrong place you will be marked wrong until you fix the problem.
$ ln /home/demo/bigfile biglink

Q8. Soft link
Create a soft link in progs called mylink, which is a relative soft link to the thismonth file in your home directory.
$ ln -s ../../thismonth mylink

Q9. Soft link – absolute
Create a soft link in progs called mylink2, which is an absolute soft link to the thismonth file in your home directory.
$ ln -s /home/demo/thismonth mylink2

Permissions

Q2. Octal code
What is the octal value for the permissions on /home/demo/bigfile?
$ ls -l bigfile
Octal permissions = 0644

Q3. Other permissions
What is the “other” permissions on /home/demo/bigfile?
$ ls -l bigfile
Other permissions = Read

Q4. Remove read and execute
Take away from the scripts directory read and execute (search) permission for yourself only.
$ ls -ld scripts
$ chmod u-rx scripts

Q5. Impact of no rx
What did removing read and execute do to your ability to look into scripts?
I could no longer ls the directory.

Q6. Add read
Add in read access for yourself on the scripts directory.
$ chmod u+r scripts

Q7. Impact of no x
What did restoring read access do to your ability to look into scripts?
I could ls the directory, but not access the contents.

Q8. Add execute
Restore the directory scripts to read, write, execute for the owner, and read and execute for both others and group.
$ chmod u+x scripts

Q9. umask
What umask setting would be required to give, when creating a directory, read/write/execute for yourself, read and execute for the group, but only execute for others?
Umask = 026

To determine the umask value you want to set, subtract the value of the permissions you want from 666 (for a file) and 777 (for a directory). The remainder is the value to use with the umask command.

On Linux and other Unix-like operating systems, new files are created with a default set of permissions.
Specifically, a new file's permissions may be restricted in a specific way by applying a permissions "mask" called the umask. The umask command is used to set this mask, or to show you its current value.

Pipes

Q2. sort
Using cat create a pipe that will concatenate the two files club_members and names, sort them and send the output to the file s1.
$ cat club_members names | sort > s1

Q3. reverse sort
Using cat create a pipe that will concatenate the two files club_members and names, sort them in reverse order and send the output to the file s2.
$ cat club_members names | sort > s2

Q4. sort on a field
Sort the file club_members numerically by meetings attended (this is the last filed). Send the output to file s3.
$ sort -k 5n club_members > s3

Q5. sort with field separator
Sort the /etc/passwd numerically by user id numbers (third field) in ascending order. You may wish to use –t option here. Send the output to s4
$ sort -n -t ':' -k3 /etc/passwd > s4

Q6. using grep
Use grep on /usr/share/dict/words to find the first word that contains the three-letter sequence 'wta'.
$ grep -E 'wta' /usr/share/dict/words | head -1

Q7. count with grep
Use grep piped through wc on /usr/share/dict/words to find the number of words that contain the letter x.
$ grep -E 'x' /usr/share/dict/words | wc -w

Q8. negative grep
Use grep to find all names in /etc/passwd that do not nave nologin on the line. Make grep include the line numbers of the matching lines and send the output to file s5.
$ grep -E -v -n 'nologin' /etc/passwd > s5 (-v: invert match)

Q9. grep and ls
Use ls –l and grep to find all the files in the directory /etc that were last modified in August (Hint: try looking for the case sensitive string "Aug").
Send this list to s6.
$ find /etc -type f -name 'Aug' > s6 File empty but shows correct in linuxzoo ???
$ ls-l /etc | grep 'Aug' – Proper command, linuxzoo wrong.

Q10. ls grep and sort
Use ls –l and grep and sort to find all the files in /etc that were last modified in Jun. Sort this list in descending order of size and then alphabetically by name (so 2 files with the same size will appear in alphabetical order).

Send the output to s7. Sorting using other techniques will probably not get the same answer.
$ ls –l /etc | grep 'Jun' | sort –k5rn –k9 > s7

Q11. find root files
Find all the files and directories in /var (including subdirectories) that are owned by user root. Send the list of full path names to s8.

Your find command may produce "Permission Denied" errors during this search. This will have no effect on the answer, and this error can be safely ignored.
$ find /var | -user root > s8

Q12. find .conf files
Find all the files in /etc (including subdirectories) ending in .conf. Send the list of full path names to s9.

Your find command may produce "Permission Denied" errors during this search. This will have no effect on the answer, and this error can be safely ignored.
$ find /etc -name '*.conf' > s9

Q13. find new files
Find all the files and directories in the demo directory that are newer than s1. Send the output of the command to /var/tmp/t1 (don't send it to the demo directory). The names of the files should appear as full names.

For example, the file "s5" would appear as "/home/demo/s5". The "/home/demo", if it appears in the output, should not have a trailing "/". The secret to avoiding the trailing slash is to use "/home/demo" not "/home/demo/" in the find command.
$ find /home/demo –newer s1 > /var/tmp/t1

Q14. list large files
Find all the files in the directory /etc/ and its subdirectories that are larger than 1 megabyte (which you can assume is 2048 blocks of 512 bytes). Send the output to s10.

Your find command may produce "Permission Denied" errors during this search. This will have no effect on the answer, and this error can be safely ignored.
$ find /etc/ -size +2048 > s10

Q15. small xml files
Create a directory called smallc in /home/demo. Copy into it all the files that begin with s from /usr/include that are smaller or equal to 12K.

Your find command may produce “Permission Denied” errors during this search. This will have no effect on the answer, and this error can be safely ignored.
$ find /usr/include –name ‘s*’ –size -12001c –exec cp {} /home/demo/smallc \;

Essential Unix commands

We explore some of the basic unix commands such as ls, ps, chmod. From now on you must be logged in as root, not demo.

Q1. ls
Enter the number of entries in the top-level directory tree. You can use the command ls / and simply count the words.
If your too last to count the words, you can pass the output through the wc command: ls / | wc
$ ls / | wc -w

Number of files = 20

Q2. edit
Edit the file /etc/motd so that it contains the single word Welcome. You can use the echo command and redirect the output, or you can use an editor such as vi or nano (similar to pico). If you get a permission denied error ARE YOU SURE YOU ARE ROOT?

On some Windows machines, nano does not like the cursor keys. If (and only if) the cursor keys result in funny characters when using nano, type the following at the prompt and then try nano again:

export TERM=vt102

This should get you going again.
$ vi /etc/motd
In vi we add the message 'Welcome'

Q3. uid
Give the uid of the user called "operator".
$ id -u operator
$ cat /etc/passwd | grep –E 'operator' – can also do it this way

uid = 11

Q4. owner
Who is the owner of the directory /var/cache/httpd? If you get no information, are you sure you are using 'ls' to give the directory information of the contents of the directory instead?
$ ls -l /var/cache/httpd

owner = apache

Q5. permissions
Give the name of the first directory (alphabetically) of / that has no read permission for other.
$ ls -l / - then look for directory. – is there another way??

Directory name = root

Q6. more permissions
Change the permission of the directory /var/log/httpd so that group and world have execute and read permissions.
$ ls –ld /var/log/httpd – to see the current permissions
$ chmod 0755 /var/log/httpd

Q7. change owner
Change the owner of the file/etc/ntp.conf to operator.
$ ls -l /etc/ntp.conf – to see the current permissions, owners etc.
$ chown operator.root /etc/ntp.conf

Q8. drop zone
Create a directory /root/dropzone. It should be set up so that group and other users can save files in the directory, but they cannot read the files that are there. User root must be able to read and write the directory.
$ mkdir /root/dropzone
$ ls -ld /root/dropzone – to see permissions of new directory before setting them.
$ chmod go-r /root/dropzone
$ chmod go+w /root/dropzone – write means you can save the file

Q9. linking files
Create a symbolic link so that the file /usr/share/dict/words appears as /root/words
$ ln -s /usr/share/dict/words /root/words

Q10. Word puzzle 1
This is a challenge question. Miss it out if you don't know regular expressions.

Use grep on words to find a word that contains each of the vowels in the correct order. How many such words are there? (you may include words with extra vowels such as adventitious).
$ cat words | grep -E 'a.*e.*i.*o.*u' | wc -w

Words = 247

Q11. Word puzzle 2
This is a challenge question. Miss it out if you don't know regular expressions.

The word minglingly includes the same four characters (ingl) repeated. How many such words are there which also begin with lower case 'm' (with any four characters repeated).
$ grep -E '^m.*(….).*\1' /usr/share/dict/words | wc -w

Words = 27

Q12. How much space…?
Look in /usr/share/doc and find a directory listing starting 'git-' followed by a version number. In the following questions this directory is referred to as the 'git' directory.

How much space is being used by the git directory? Use a command to calculate this, and don't try adding it up yourself. We want the total answer in human readable format (e.g.6.2M).
$ ls -l /usr/share/doc | grep -E 'git'
$ du -sh /usr/share/doc/git-1.8.3.1

Space = 12M

Q13. How much space is available?
In human readable form (e.g.123M) how much disk space is available on the main filesystem?
$ df -h / - '-h' flag puts sizes in human readable form (e.g. G/M rather than blocks)

Space = 2.2G

Q14. How much space is available?
Remove the entire git directory (it is not used in our tutorials). Now repeat the above calculation for disk space, but this time give the answer in blocks.
$ df – no flag, shows the sizes in blocks

Space = 2233120

Essential Unix Administration

Q1. Partitions and LVM
Use sfdisk with block units and find out the partitions which exist in /dev/sda. How many blocks are in the first partition?
$ sfdisk -l /dev/sda

Blocks = 512000

Use the pvdisplay command of LVM to discover what physical volume (i.e. which partition) is being managed by LVM. What is the partition being used (PV Name) and what is the volume group name (VG Name)?
$ pvdisplay

PV Name = /dev/sda2
VG Name = centos_lvm

Use lvdisplay to discover information about the VG Name found in the previous question. What is the first LV Path which is using the volume group discovered in the previous question?
$ lvdisplay

LV Path = /dev/centos_lvm/root

Using the path discovered in the previous question, look at this path in the /dev directory using a long listing ls command. Assuming this is in fact a softlink, what is the ABSOLUTE device name which this link is pointing to?
$ cd /dev
$ ls –l /dev/centos_lvm/root

Absolute device name = /dev/dm-0

From mounting this logical volume, the current method is NOT to use the volume name, or even the device it points to. Instead the device mapper is used, which can support different layers (such as encryption on top of something else). This can be found in /dev/mapper.

Look in /dev/mapper and find the softlink which points to the device file identified in the previous question. What is the relative name of this link? So if the link was '/dev/mapper/gordon', the answer wanted here is 'gordon'.
$ ls –l /dev/mapper

Mapper link name = centos_lvm-root

Look in the fstab mount table. Find the line which mounts this partition via the mapper device. Where is this partition mounted?
$cat /etc/fstab

Mount directory = /

One can also mount things using the filesystem block id (which is the UUID shown in fstab). What block id could you use instead of the mapper mount in this case?
It is in a format like ‘ffffff-ffff-fffff-fffff-fffffffff’
$ blkid

Block id for mapper filesystem = b66fdf9b-16f0-4648-9663-536881db0ab1

Recall that you discovered the device file in /dev which the LVM mapper entry was soft linked to. What was the major and minor numbers of this device?
$ ls –l /dev | grep –E ‘dm-0’

Major number = 253
Minor number = 0

Somewhere in the /proc filesystem there is a file which tells you how much swap space has been allocated to the computer. Find that file and then find out how big in bytes the swap space is. Hint: the information that you require is located within the /proc directory within a file.
$ free

Enter a number = 2103484

Q2. Processes and Services
What is the process id of rsyslogd? Hint: remember the ‘ps aux’ command.
$ ps aux | grep -E ‘rsyslog’

Enter a number = 1004

Kill rsyslogd using the kill command.
$ kill -s SIGKILL 1004
$ kill 1004 – what is the difference

Using systemctl, get the status of the rsyslog service. What is the full path to the system configuration file which controls the rsyslog service?
$ systemctl status rsyslog

Systemd file = /usr/lib/systemd/system/rsyslog.service

Look at this configuration file. Find the line which configures the environmental variables of rsyslog (Environment File). Ignoring the '=' or the '=-' if it exists, what is the environment file for this service?
$ cat /usr/lib/systemd/system/rsyslog.service

Config file = /etc/sysconfig/rsyslog

Restart the rsyslogd services using systemctl. Confirm this is running using the status option.
$ systemctl restart rsyslog
$ systemctl status rsyslog

$ service rsyslog start – would this also work?
$ service rsyslog status – would this also work?

Q3. Control a service
Start the database. This is called mariadb. This may take a few seconds.
$ service mariadb start

Using systemctl, discover the process id (PID) for the main process. Note, mariadb has many processes, so make sure you select the Main PID.
$ systemctl status mariadb

Main PID = 3802

What user is the owner of this process? Use the 'ps' command with the appropriate flags.
$ pa aux | grep '3802'

Main PID owner = mysql

If the main pid is the parent of the database, what is the PID of the first child of this parent pid? So, if you have the process name for this PID, then pstree might help here.
$ pstree –p 3344

PID of First Child = 3502

Set the mariadb to run next time you boot your virtual machine.
$ systemctl enable mariadb.service

Systemctl has an option 'list-unit-files', which says which things are enabled for running at boot. How many units are enabled?
$ systemctl list-unit-files | grep –E 'enabled' | wc -l

Units = 73

How many enabled units are socket units?
$ systemctl list-unit-files --type=socket | grep –E ‘enabled’ | wc -l

Sockets = 10

Now set the mariadb so that it DOES NOT run next time you boot your virtual machine.
$ systemctl disable mariadb.service

Linux Networking

Question 2. Main Network

What is the network device name which connects you to the linuxzoo network? This wil have an IPv4 address which starts “10.”.. Do not type in /dev when you answer this (i.e. make that answer relative to /dev).

$ ip addr show

```
[root@host-0-0 ~]# ip link show lo
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
[root@host-0-0 ~]# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 00:0c:49:00:06:90 brd ff:ff:ff:ff:ff:ff
    inet 10.0.6.73/29 brd 10.0.6.79 scope global dynamic ens3
       valid_lft 863732sec preferred_lft 863732sec
    inet6 fe80::9d33:d8b:c4fc:88/64 scope link
       valid_lft forever preferred_lft forever
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN qlen 1000
    link/ether 52:54:00:fb:c1:2c brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
       valid_lft forever preferred_lft forever
4: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc pfifo_fast master virbr0 state DOWN qlen 1000
    link/ether 52:54:00:fb:c1:2c brd ff:ff:ff:ff:ff:ff
5: eth2@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen 1000
    link/ether e2:f5:fb:78:96:ef brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::e0f5:fbff:fe78:96ef/64 scope link
       valid_lft forever preferred_lft forever
6: eth3@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen 1000
    link/ether 4a:82:54:a2:aa:e3 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::4882:54ff:fea2:aae3/64 scope link
       valid_lft forever preferred_lft forever
[root@host-0-0 ~]# ip addr show en3
Device "en3" does not exist.
[root@host-0-0 ~]# ip addr show ens3
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 00:0c:49:00:06:90 brd ff:ff:ff:ff:ff:ff
    inet 10.0.6.73/29 brd 10.0.6.79 scope global dynamic ens3
       valid_lft 863157sec preferred_lft 863157sec
    inet6 fe80::9d33:d8b:c4fc:88/64 scope link
       valid_lft forever preferred_lft forever
```

Main network device = ens3

What is the broadcast address for this network connection?

$ used the same command as above

Enter an IP = 10.0.6.239

What is the netmask in quad dotted format for this network connection?

$ ifconfig – is there another way we should’ve done this?

Enter an IP 00.00.00.00 = 255.255.255.248

What is the IP for the default route (i.e.the gateway) for your virtual machine?
$ ip route

```
[root@host-0-0 ~]# ip route
default via 10.0.6.78 dev ens3  proto static  metric 100
10.0.6.72/29 dev ens3  proto kernel  scope link  src 10.0.6.73  metric 100
192.168.122.0/24 dev virbr0  proto kernel  scope link  src 192.168.122.1
[root@host-0-0 ~]#
```

IP number = 10.0.6.78

Device eth2 is connected to a network for which your virtual machine is the gateway. The network IP is 192.168.1.0 and the netmask is 255.255.255.0. Your IP should be the last valid host IP number allowed for that network.
I worked this out using pen and pad. IP address to binary, split using netmask, get broadcast address, work out 1st and last usable host.
Calculate your IP/NETWORK configuration for eth3.

IP number 00.00.00.00/32 = 192.168.1.254/24

Configure eth2 as per the previous question. Confirm operations by pinging 192.168.1.23. Make sure the broadcast address is correct too.
$ ip address add 192.168.1.254/24 broadcast 192.168.1.255 dev eth2
$ ping 192.168.1.23 – pings worked, sent reply.

Device eth3 is connected to a network for which your virtual machine is the gateway. The network IP is 192.168.3.48 and the netmask is 255.255.255.240. Your IP should be the last valid host IP number allowed for that network.
I worked this out using pen and pad. IP address to binary, split using netmask, get broadcast address, work out 1st and last usable host.
Calculate your IP/NETWORK configuration for eth3.

IP number 00.00.00.00/32 = 192.168.3.62/28

Configure eth3 as per the previous question. Confirm operations by pinging 192.168.3.50. Make sure the broadcast address is correct too.
$ ip address add 192.168.3.62/28 broadcast 192.168.3.63 dev eth3
$ ping 192.168.3.50 – pings worked, sent reply

Using an “IP link” command, discover the mac address of eth3

$ ip link

```
[root@host-0-0 ~]# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP mode DEFAULT qlen 1000
    link/ether 00:0c:49:00:06:90 brd ff:ff:ff:ff:ff:ff
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT qlen 100
0
    link/ether 52:54:00:fb:c1:2c brd ff:ff:ff:ff:ff:ff
4: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc pfifo_fast master virbr0 state DOWN mode DEFAULT q
len 1000
    link/ether 52:54:00:fb:c1:2c brd ff:ff:ff:ff:ff:ff
5: eth2@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT qlen 1000
    link/ether e2:f5:fb:78:96:ef brd ff:ff:ff:ff:ff:ff link-netnsid 0
6: eth3@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT qlen 1000
    link/ether 4a:82:54:a2:aa:e3 brd ff:ff:ff:ff:ff:ff link-netnsid 1
[root@host-0-0 ~]#
```

Mac aa:aa:aa:aa:aa:aa = 4a:82:54:a2:aa:e3

What is the mac address of 192.168.3.50? Hint: “/proc”

$ arp -a

If you were able to log onto 192.168.3.50, how would you have configured the default route?

Worked out by hand, one less than broadcast address.

ip route add default via 192.168.3.62 dev eth0

Question 3. Listening services and connections

How many programs are listening on IPv4 sockets on your machine?

Hint: the netstat command will help you here.

$ netstat –lt

```
[root@host-6-233 ~]# netstat -lt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:sunrpc          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:38704           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:mountd          0.0.0.0:*               LISTEN
tcp        0      0 host-0-0.linuxzo:domain 0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 localhost:smtp          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:nfs             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:54571           0.0.0.0:*               LISTEN
tcp6       0      0 [::]:sunrpc             [::]:*                  LISTEN
tcp6       0      0 [::]:mountd             [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
tcp6       0      0 [::]:telnet             [::]:*                  LISTEN
tcp6       0      0 localhost:smtp          [::]:*                  LISTEN
tcp6       0      0 [::]:59326              [::]:*                  LISTEN
tcp6       0      0 [::]:nfs                [::]:*                  LISTEN
tcp6       0      0 [::]:36265              [::]:*                  LISTEN
[root@host-6-233 ~]#
```

Only count tcp connections, not tcp6.

Number of TCP listeners = 8

What is the PID of the process which is listening on UDP port 111 (sunrpc).
Hint: the netstat command will help you here.
$ netstat –tulpn

```
udp        0      0 0.0.0.0:67              0.0.0.0:*                           1246/dnsmasq
udp        0      0 127.0.0.1:323           0.0.0.0:*                           615/chronyd
udp        0      0 0.0.0.0:68              0.0.0.0:*                           761/dhclient
udp        0      0 0.0.0.0:20048           0.0.0.0:*                           1054/rpc.mountd
udp        0      0 0.0.0.0:111             0.0.0.0:*                           650/rpcbind
udp        0      0 0.0.0.0:35490           0.0.0.0:*                           -
udp6       0      0 :::2049                 :::*                                -
udp6       0      0 :::817                  :::*                                650/rpcbind
udp6       0      0 :::52031                :::*                                -
udp6       0      0 ::1:323                 :::*                                615/chronyd
udp6       0      0 :::20048                :::*                                1054/rpc.mountd
udp6       0      0 :::111                  :::*                                650/rpcbind
udp6       0      0 :::41848                :::*                                1014/rpc.statd
udp6       0      0 :::6586                 :::*                                761/dhclient
[root@host-0-0 ~]#
```

PID of listener = 650

When you connected to linuxzoo.net via telnet or ssh, it was forwarded from 10.200.0.1 (the main linuxzoo server) to your machine's telnet or ssh server using a proxy.
Find the port number on the 10.200.0.1 end of one of these proxy connections and enter it below.
Hint: again, try the netstat command very near the beginning of the output…It sometimes helps to use the -n flag, as this does not do DNS lookups.
$ netstat –n

```
[root@host-0-0 ~]# netstat -n
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 10.0.6.233:22           10.200.0.1:26857        ESTABLISHED
tcp        0      0 10.0.6.233:22           10.200.0.1:43393        ESTABLISHED
udp        0      0 10.0.6.233:54052        178.62.68.79:123        ESTABLISHED
udp        0      0 10.0.6.233:56385        178.79.145.244:123      ESTABLISHED
udp        0      0 10.0.6.233:37206        139.162.219.252:123     ESTABLISHED
udp        0      0 10.0.6.233:36787        129.250.35.251:123      ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ]         DGRAM                    11527    /run/systemd/shutdownd
unix  2      [ ]         DGRAM                    7013     /run/systemd/notify
unix  2      [ ]         DGRAM                    7015     /run/systemd/cgroups-agent
```

10.200.0.01 proxy port = 26857 0r 43393

Question 4. Traceroute: hop count

In order for your virtual machine to reach the internet, it's packets travles through a number of virtual networks. The final network node is 10.200.0.1.
Using traceroute, find out how many hops it takes to reach 10.200.0.1. Note you must use ICMP ECHO in traceroute, rather than the default. Find the right flag in the manual.
$ traceroute –I 10.200.0.1

```
[root@host-0-0 ~]# traceroute -I 10.200.0.1
traceroute to 10.200.0.1 (10.200.0.1), 30 hops max, 60 byte packets
 1  host-6-238.linuxzoo.net (10.0.6.238)  0.238 ms  0.167 ms  0.245 ms
 2  hub1-gw.linuxzoo.net (10.200.0.1)  0.423 ms  0.379 ms  0.271 ms
[root@host-0-0 ~]#
```

Num of hops = 2

Question 5. nmap: Open ports

Use nmap to analyse the ports open on 10.200.0.1. As the nmap command can take quite a while to run, restrict your scan to open tcp ports between port numbers 50 to 80 inclusive.
List the open port numbers you find with spaces between them in the box below (e.g. if ports

50 and 60 are open, the answer is “50 60”). The numbers in your list must be sorted (smallest number first).

IMPORTANT: Linuxzoo security may shut you down if you produce too many packets too quickly. Use the following options for nmap or you may be kicked off the system. Even with these options the scan may take quite a few seconds.

$ nmap 10.200.0.1 -p 50-80 --max-retries 3

```
[root@host-0-0 ~]# nmap 10.200.0.1 -p 50-80 --max-retries 3

Starting Nmap 6.40 ( http://nmap.org ) at 2019-10-18 16:10 BST
Nmap scan report for hub1-gw.linuxzoo.net (10.200.0.1)
Host is up (0.00043s latency).
Not shown: 29 filtered ports
PORT   STATE SERVICE
53/tcp open  domain
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 1.54 seconds
[root@host-0-0 ~]#
```

Open ports = 53 and 80

Question 6. tcpdump and web requests

The tcpdump command allows us to capture all or some of the network traffic on a particular network device. In this question use the tcpdump command to capture the behaviour of a web page request emulated using lwp-request.

Capturing packets successfully can be tricky, especially since you have to use the same network to talk to your machine, and you don’t want to capture that too…. Therefore, consider the following commands to perform the capturing. NOTE: you only type the characters in bold.

```
$ tcpdump -vi eth0 port 80 > /tmp/log &
[1] 3123
$ lwp-request http://linuxzoo.net
...blah...blah
...blah...blah
$ kill -1 %1
[1]+ Done ....
$
```

Where it says “eth0” you must replace that with your main network connection drive, which you discovered right at the beginning of this tutorial.

Now look at the contents of /tmp/log and enter in the box below the common IP flag which appears in most packets. (The information can be found between the [..] brackets immediately after the text “flags”. Ignore the information after the text “Flags” e.g. look for the one in all lowercase). Enter the two letters seen in the box below.

```
[root@host-0-0 ~]# cat /tmp/log
16:19:50.885894 IP (tos 0x0, ttl 64, id 57821, offset 0, flags [DF], proto TCP (6), length 60)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [S], cksum 0x49c9 (incorrect -> 0x8bd4), seq 107101973, win 29200, options [mss 1460,sackO
4316872 ecr 0,nop,wscale 6], length 0
16:19:50.886344 IP (tos 0x0, ttl 63, id 0, offset 0, flags [DF], proto TCP (6), length 60)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [S.], cksum 0x2427 (correct), seq 566553897, ack 107101974, win 28960, options [mss 1460,s
val 1839262716 ecr 4316872,nop,wscale 7], length 0
16:19:50.886360 IP (tos 0x0, ttl 64, id 57822, offset 0, flags [DF], proto TCP (6), length 52)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [.], cksum 0x49c1 (incorrect -> 0xc24a), ack 1, win 457, options [nop,nop,TS val 4316872 e
2716], length 0
16:19:50.902619 IP (tos 0x0, ttl 64, id 57823, offset 0, flags [DF], proto TCP (6), length 184)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [P.], cksum 0x4a45 (incorrect -> 0x2f95), seq 1:133, ack 1, win 457, options [nop,nop,TS v
9 ecr 1839262716], length 132
16:19:50.903035 IP (tos 0x0, ttl 63, id 295, offset 0, flags [DF], proto TCP (6), length 52)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [.], cksum 0xc282 (correct), ack 133, win 235, options [nop,nop,TS val 1839262733 ecr 4316
gth 0
16:19:51.077306 IP (tos 0x0, ttl 63, id 296, offset 0, flags [DF], proto TCP (6), length 8379)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [P.], cksum 0x6a48 (incorrect -> 0x45b9), seq 1:8328, ack 133, win 235, options [nop,nop,T
9262907 ecr 4316889], length 8327
16:19:51.077339 IP (tos 0x0, ttl 64, id 57824, offset 0, flags [DF], proto TCP (6), length 52)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [.], cksum 0x49c1 (incorrect -> 0x9ebd), ack 8328, win 717, options [nop,nop,TS val 431706
9262907], length 0
16:19:51.077401 IP (tos 0x0, ttl 63, id 302, offset 0, flags [DF], proto TCP (6), length 79)
```

What is the common IP flag seen in the log = DF

Each packet is split over multiple lines. A new packet starts on a fresh line, beginning with a timestamp. This should be followed with IP and then IP header information.
Assuming the packet is a TCP packet, the next link should include “Flags” (with a capital F) which are the TCP flags.
Find the packets involved with the TCP Fin flag. The first of those is the packet which first requested the stream be closed. What is the TCP sequence number of that packet?
$ cat /tmp/log

```
[root@host-0-0 ~]# cat /tmp/log
16:19:50.885894 IP (tos 0x0, ttl 64, id 57821, offset 0, flags [DF], proto TCP (6), length 60)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [S], cksum 0x49c9 (incorrect -> 0x8bd4), seq 107101973, win 29200, options [mss 1460,sackOK,TS val
4316872 ecr 0,nop,wscale 6], length 0
16:19:50.886344 IP (tos 0x0, ttl 63, id 0, offset 0, flags [DF], proto TCP (6), length 60)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [S.], cksum 0x2427 (correct), seq 566553897, ack 107101974, win 28960, options [mss 1460,sackOK,TS
val 1839262716 ecr 4316872,nop,wscale 7], length 0
16:19:50.886360 IP (tos 0x0, ttl 64, id 57822, offset 0, flags [DF], proto TCP (6), length 52)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [.], cksum 0x49c1 (incorrect -> 0xc24a), ack 1, win 457, options [nop,nop,TS val 4316872 ecr 183926
2716], length 0
16:19:50.902619 IP (tos 0x0, ttl 64, id 57823, offset 0, flags [DF], proto TCP (6), length 184)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [P.], cksum 0x4a45 (incorrect -> 0x2f95), seq 1:133, ack 1, win 457, options [nop,nop,TS val 431688
9 ecr 1839262716], length 132
16:19:50.903035 IP (tos 0x0, ttl 63, id 295, offset 0, flags [DF], proto TCP (6), length 52)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [.], cksum 0xc282 (correct), ack 133, win 235, options [nop,nop,TS val 1839262733 ecr 4316889], len
gth 0
16:19:51.077306 IP (tos 0x0, ttl 63, id 296, offset 0, flags [DF], proto TCP (6), length 8379)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [P.], cksum 0x6a48 (incorrect -> 0x45b9), seq 1:8328, ack 133, win 235, options [nop,nop,TS val 183
9262907 ecr 4316889], length 8327
16:19:51.077339 IP (tos 0x0, ttl 64, id 57824, offset 0, flags [DF], proto TCP (6), length 52)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [.], cksum 0x49c1 (incorrect -> 0x9ebd), ack 8328, win 717, options [nop,nop,TS val 4317063 ecr 183
9262907], length 0
16:19:51.077401 IP (tos 0x0, ttl 63, id 302, offset 0, flags [DF], proto TCP (6), length 79)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [P.], cksum 0xd6f3 (correct), seq 8328:8355, ack 133, win 235, options [nop,nop,TS val 1839262907 e
cr 4316889], length 27
16:19:51.077413 IP (tos 0x0, ttl 64, id 57825, offset 0, flags [DF], proto TCP (6), length 52)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [.], cksum 0x49c1 (incorrect -> 0x9ea2), ack 8355, win 717, options [nop,nop,TS val 4317063 ecr 183
9262907], length 0
16:19:51.081415 IP (tos 0x0, ttl 63, id 303, offset 0, flags [DF], proto TCP (6), length 52)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [F.], cksum 0xa07f (correct), seq 8355, ack 133, win 235, options [nop,nop,TS val 1839262911 ecr 43
17063], length 0
16:19:51.087538 IP (tos 0x0, ttl 64, id 57826, offset 0, flags [DF], proto TCP (6), length 52)
    host-6-233.linuxzoo.net.56240 > activesql.soc.napier.ac.uk.http: Flags [F.], cksum 0x49c1 (incorrect -> 0x9e91), seq 133, ack 8356, win 717, options [nop,nop,TS val 43170
74 ecr 1839262911], length 0
16:19:51.087882 IP (tos 0x0, ttl 63, id 304, offset 0, flags [DF], proto TCP (6), length 52)
    activesql.soc.napier.ac.uk.http > host-6-233.linuxzoo.net.56240: Flags [.], cksum 0xa06d (correct), ack 134, win 235, options [nop,nop,TS val 1839262917 ecr 4317074], len
gth 0
```

Seq no = 8355

SELinux Administration

Question 1: Global Settings

Use the getenforce administrative command. What is the current setting?
$ getenforce

Setting=Enforcing

What is the absolute pathname to the SELinux directory in /sys?
$ find / -type d -name selinux

Pathname=/sys/fs/selinux

How does the information from getenforce compare to the related enforce status value stored in /sys?
$ cat /sys/fs/selinux/enforce
It is stored as the value “1”, rather than enforce. The value is always stored as a “1” or a “0”.

How many files and directories are actually in the top level of the SELinux directory in /sys?
$ cd /sys/fs/selinux
$ ls -Z | wc -l

Number of files=23

Question 2: Basic Labels

Locate the syslog daemon (called rsyslogd). What is the full true pathname?
$ ps auxZ | grep rsyslogd

```
[root@host-6-73 ~]# ps auxZ | grep rsyslogd
system_u:system_r:syslogd_t:s0  root      1001  0.0  0.2 215680  1436 ?        Ssl  10:38   0:00 /usr/sbin/rsyslogd -n
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 root 1893 0.0  0.1 112648 920 pts/0 R+ 10:57   0:00 grep --color=auto rsyslogd
[root@host-6-73 ~]#
```

Pathname = /usr/sbin/rsyslogd

What is the SELinux label of this executable rsyslogd file?
$ ls -Z /usr/sbin/rsyslogd

```
[root@host-6-73 ~]# ls -Z /usr/sbin/rsyslogd
-rwxr-xr-x. root root system_u:object_r:syslogd_exec_t:s0 /usr/sbin/rsyslogd
[root@host-6-73 ~]#
```

User= system_u
Role= object_r
Type= syslogd_exec_t
Sensitivity= s0

The daemon rsyslogd uses /etc/rsyslog.conf as its configuration file. What is the SELinux label of the rsyslogd configuration file?

$ ls -Z /etc/rsyslog.conf

```
[root@host-6-73 ~]# ls -Z /etc/rsyslog.conf
-rw-r--r--. root root system_u:object_r:syslog_conf_t:s0 /etc/rsyslog.conf
[root@host-6-73 ~]#
```

User= system_u
Role= object_r
Type= syslog_conf_t
Sensitivity= s0

Given that rsyslogd is running currently, what is the label of the process. Use the list of running processes to discover this.

$ ps auxZ | grep rsyslogd

```
[root@host-6-73 ~]# ps auxZ | grep rsyslogd
system_u:system_r:syslogd_t:s0  root      1000  0.0  0.4 215680  2080 ?        Ssl  11:29   0:00 /usr/sbin/rsyslogd -n
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 root 1901 0.0  0.1 112648 924 pts/0 R+ 11:43   0:00 grep --color=auto rsyslogd
[root@host-6-73 ~]#
```

User= system_u
Role= system_r
Type= syslogd_t
Sensitivity= s0

With the label of the running process, and the label of the configuration file, use sesearch to find the semantic rules to allow the process to read the configuration file. Make sure you look only for allow rules and limit the search to the specific source type and target type, and also limit your search to just file rules.
Save the output of this command to /root/selinux1.

$ sesearch --allow --source syslogd_t --target syslog_conf_t --class file > /root/selinux1

There are other allow rules, not just ones which relate to files. Confirm the existence of one for accessing directories of the syslog daemon's configuration file label. Use a class of "dir" to do this.
Again, there should only be 1.
Save the output of this command to /root/selinux2 and confirm the contents visually.

$ sesearch --allow --source syslogd_t --target syslog_conf_t --class dir > /root/selinux2

What directories in the top level of /etc have this configuration type label?
Use an "ls -Z" command on /etc and combine it with 2 greps so you locate the correct syslog configuration label while restricting your search to just directories.
Save this output to /root/selinux3.

$ ls -Z /etc | grep syslog_conf_t | grep .d > /root/selinux3

Now use the “find” command to find all files and directories in /etc which have this configuration type label. You need to use -context.
HINT: -context is the whole label, so use filename-style wildcards so you only need to specify the type.
Save this output to /root/selinux4.
$ find /etc -context '*syslog_conf_t*' > /root/selinux4

Question 3: Port Rules

The syslog daemon you investigated is allowed to open a number of ports, both TCP and UDP. Use the sesearch on the syslogd_t type, focusing on TCP sockets and the name_bind permission. Include the -C to better understand conditional rules.
You should ignore rules where the line begins with DT or DF. This indicates the conditional rule is currently disabled.
How many ENABLED name_bind permissions are allowed from syslogd_t.
$ sesearch -s syslogd_t -c tcp_socket -AC -p name_bind

```
[root@host-6-73 ~]# sesearch -s syslogd_t -c tcp_socket -AC -p name_bind
Found 6 semantic av rules:
   allow syslogd_t syslogd_port_t : tcp_socket { name_bind name_connect } ;
   allow syslogd_t syslog_tls_port_t : tcp_socket { name_bind name_connect } ;
   allow syslogd_t rsh_port_t : tcp_socket { name_bind name_connect } ;
DT allow nsswitch_domain port_t : tcp_socket name_bind ; [ nis_enabled ]
DT allow nsswitch_domain ephemeral_port_t : tcp_socket name_bind ; [ nis_enabled ]
DT allow nsswitch_domain unreserved_port_t : tcp_socket name_bind ; [ nis_enabled ]
```

Name_bind permissions= 3

Use the types of the previous answer to loop up the ports associated with those types. Taking ONLY the TCP ports, make a list, ordered in ascending port number, separated by commas if necessary, and without any white space.
So, if all the port types together give you TCP ports 1, 5 and 10, your answer would be “1,5,10”.
$ semanage port -l | grep -e syslogd_port_t -e syslog_tls_port_t -e rsh_port_t | grep tcp | sort -n

```
[root@host-6-73 ~]# semanage port -l | grep -e syslogd_port_t -e syslog_tls_port_t -e rsh_port_t | grep tcp | sort -n
rsh_port_t                     tcp      514
syslogd_port_t                 tcp      601
syslog_tls_port_t              tcp      6514
[root@host-6-73 ~]#
```

Ports= 514,601,6514

Take the highest numbered TCP port you discovered from the last question and look it up in /etc/services. What is the service name (the first column) for this port?
$ cat /etc/services | grep 6514

```
[root@host-6-73 ~]# cat /etc/services | grep 6514
syslog-tls      6514/tcp                # Syslog over TLS
syslog-tls      6514/udp                # Syslog over TLS
syslog-tls      6514/dccp               # Syslog over TLS
```

Service name= syslog-tls

Question 4: Process Transitions

Locate the Network Manager daemon (called NetworkManager). What is the full true pathname?

$ find / -name NetworkManager

```
[root@host-6-73 ~]# find / -name NetworkManager
/run/NetworkManager
/etc/NetworkManager
/var/lib/NetworkManager
/usr/sbin/NetworkManager
/usr/lib/NetworkManager
/usr/lib64/NetworkManager
/usr/share/doc/NetworkManager
```

Given that NetworkManager is running currently, what is the label of the process. Use the list of running processes to discover this.

$ ps auxZ | grep /usr/sbin/NetworkManager

```
[root@host-6-73 ~]# ps auxZ | grep /usr/sbin/NetworkManager
system_u:system_r:NetworkManager_t:s0 root 727  0.0  0.3 525396  1924 ?        Ssl  12:31   0:00 /usr/sbin/NetworkManager --no-daemon
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 root 2099 0.0  0.1 112648 928 pts/0 R+ 12:54   0:00 grep --color=auto /usr/sbin/NetworkManager
[root@host-6-73 ~]#
[root@host-6-73 ~]#
```

User= system_u
Role= system_r
Type= NetworkManager_t
Sensitivity= s0

When NetworkManager runs, it executes files in /etc/NetworkManager/dispatcher.d whenever a network interface changes state. What is the label for the executables in the dispatcher.d directory?

$ ls -Z /etc/NetworkManager/dispatcher.d

```
[root@host-6-73 ~]# ls -Z /etc/NetworkManager/dispatcher.d
-rwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 00-netreport
-rwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 04-iscsi
-rwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 11-dhclient
-rwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 20-chrony
-rwxr-xr-x. root root unconfined_u:object_r:NetworkManager_initrc_exec_t:s0 mydns
drwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 no-wait.d
drwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 pre-down.d
drwxr-xr-x. root root system_u:object_r:NetworkManager_initrc_exec_t:s0 pre-up.d
[root@host-6-73 ~]#
```

User= system_u
Role= object_r
Type= NetworkManager_initrc_exec_t
Sensitivity= s0

When the NetworkManager process type executes a file in the dispatcher.d directory, what process transition is followed?
Find the one process transition which manages this. What process type do these files run as?

$ sesearch –all | grep NetworkManager_initrc_t

```
[root@host-6-161 dispatcher.d]# sesearch --all | grep NetworkManager_initrc_exec_t
ERROR: Cannot get avrules: Neverallow rules requested but not available
   type_transition rpm_t NetworkManager_initrc_exec_t : process initrc_t;
   type_transition init_t NetworkManager_initrc_exec_t : process initrc_t;
   type_transition certmonger_unconfined_t NetworkManager_initrc_exec_t : process initrc_t;
   type_transition initrc_t NetworkManager_initrc_exec_t : process initrc_t;
   type_transition zabbix_script_t NetworkManager_initrc_exec_t : process initrc_t;
   type_transition anaconda_t NetworkManager_initrc_exec_t : process initrc_t;
```

Process type= initrc_t

SELinux Usage

Question 1: Basic Labelling

Create 2 directories in /root, “secure” and “protect”. Set the SELinux type of secure to system_conf_t and set the type of protect to etc_t.

$ mkdir secure protect

$ ls -dZ secure – to see current type of directory

```
[root@host-6-73 ~]# ls -dZ secure
drwxr-xr-x. root root unconfined_u:object_r:admin_home_t:s0 secure
```

$ ls -dZ protect – to see current type of directory

$ chcon -t system_conf_t secure

$ ls -dZ secure

```
[root@host-6-73 ~]# ls -dZ secure
drwxr-xr-x. root root unconfined_u:object_r:system_conf_t:s0 secure
```

$ chcon -t etc_t protect

Create a file called “test1” in secure and “test2” in protect. Look at the types of these files. How does the types of these new files get decided?

$ touch secure/test1

$ touch protect/test2

$ cd secure

$ ls -Z

```
[root@host-6-73 secure]# ls -Z
-rw-r--r--. root root unconfined_u:object_r:system_conf_t:s0 test1
```

The file types are inherited form the parents.

Copy test1 to protect/test3. What happens to the test3 type in comparison to test1?

$ cp secure/test1 protect/test3

$ ls -Z protect/tets3

```
[root@host-6-73 ~]# ls -Z protect/test3
-rw-r--r--. root root unconfined_u:object_r:etc_t:s0   protect/test3
```

The type of test3 is taken from protect.

Rename secure/test1 to protect/test4. What happens to the test4 type in comparison to the type test1 was when it was in secure (system_conf_t)?

$ mv secure/test1 protect/test4

$ ls -Z protect/test4

```
[root@host-6-73 ~]# ls -Z protect/test4
-rw-r--r--. root root unconfined_u:object_r:system_conf_t:s0 protect/t
est4
```

The type of test1 is moved to test4.

Use matchpathcon to find the type which would be set if you did a restorecon on protect/test2. Save the output of matchpathcon to /root/match1.
What type would be set if you did do restorecon?
$ matchpathcon protect/test2
$ matchpathcon protect/test2 > /root/match1

```
[root@host-6-73 ~]# matchpathcon protect/test2
protect/test2   system_u:object_r:admin_home_t:s0
[root@host-6-73 ~]# matchpathcon protect/test2 > /root/match1
```

If you did a restorecon the type would be admin_home_t

Use semanage and list all of the fcontext entries, grepping the list for those which start with /root. Grep through this with the restorecon type from the previous question.
This should reduce the list to just 1 regular expression, i.e. the one which matchpathcon used to produce the answer above.
What is the expression?
$ semanage fcontext -l | grep -e '/root*' | grep admin_home_t

```
[root@host-6-73 ~]# semanage fcontext -l | grep -e '/root*' | grep admin_home_t
/root(/.*)?                                        all files          system_u:object_r:admin_home_t:s0
[root@host-6-73 ~]#
```

The expression used = /root(/.*)?

Add a rule to semanage fcontext so that any files in /root/ which end with .bin will be set to type bin_t. Create a file /root/test.bin and do a restorecon on that file to confirm it takes on bin_t.
$ semanage fcontext -a -t bin_t "/root/(.*)\.bin"
$ touch /root/test.bin
$ restorecon -v /root/test.bin

```
[root@host-6-161 ~]# restorecon -v /root/test.bin
restorecon reset /root/test.bin context unconfined_u:object_r:admin_home_t:s0->unconfined_u:object_r:bin_t:s0
[root@host-6-161 ~]# ^C
```

Question 2: Boolean control

There is a Boolean called httpd_tmp_exec. Is the Boolean on or off?
$ getsebool -a | grep httpd_tmp_exec

```
[root@host-6-73 ~]# getsebool -a | grep httpd_tmp_exec
httpd_tmp_exec --> off
```

The Boolean is off.

Change the Boolean called httpd_tmp_exec to on.
$ setsebool httpd_tmp_exec 1
$ getsebool httpd_tmp_exec – shows the status of the Boolean.

```
[root@host-6-73 ~]# getsebool httpd_tmp_exec
httpd_tmp_exec --> on
```

Find out all allow rules which are switched on by setting this boolean to on. Save the output of sesearch to /root/boolrule. When you search, find all rules, unrestricted by source types.

Question 3: Auditing

First, we click on a button to cause a mislabelling error for httpd.
Done

Start http with systemctl. It should fail…….this should cause an event.
$ systemctl start httpd

Save the AVC event to /root/event. Make sure only the AVC event is saved, and there is only 1 AVC line.
$ ausearch –m avc –ts recent | grep avc > /root/event

Use the inode information from the event. What is the full pathname of the directory in the event?
$ find / -xdev -inum 860305

Pathname = /etc/httpd/conf.d

Use restorecon on that single directory to fix the label. Confirm that httpd now starts.
$ restorecon –v /etc/httpd/conf.d
$ systemctl start httpd

Firewalls

Question 1: Firewall: Empty the Chains

In this tutorial we are going to work on the firewall configuration of your machine. Some care must be taken when doing this, or you will suddenly find you can no longer log in.
In all these cases the easiest way to do the experiment is to create an executable program in /root called “firewall”. You should make the contents something like the following:

```
#!/bin/bash
#
iptables -F INPUT
iptables -F OUTPUT
iptables -F FORWARD
iptables -P INPUT ACCEPT
iptables -P OUTPUT ACCEPT
iptables -P FORWARD DROP
#
# Accept ongoing connections
iptables -A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
#
# For your own safety, stop users logging in from other VMs
#
iptables -A INPUT -m conntrack --ctstate NEW -m tcp -p tcp --dport 22 ! -s
10.0.0.0/16 -j ACCEPT
iptables -A INPUT -m conntrack --ctstate NEW -m tcp -p tcp --dport 23 ! -s
10.0.0.0/16 -j ACCEPT

#
# Your changes go after here.
#
```

To execute the file, “chmod +x ./firewall” and then run it by using “./firewall” or “/root/firewall” to run it.
After executing this file, you can use iptables -L to show what rules have been stored in the kernel. The provided rule uses a default policy of ACCEPT, just in case you make a typo and lock yourself out. Default ACCEPT is a bad idea here, and this is just to get you going.
I copied the above text into nano and ran it. Everything worked fine.

Make the default policy of the INPUT chain DROP. Leave the other chains as they are. Do this by editing the script appropriately, then running the script. Just set the INPUT chain default once in the script.
iptables -P INPUT DROP – the below screenshot shows the chain to change.

```
#!/bin/bash
#
iptables -F INPUT
iptables -F OUTPUT
iptables -F FORWARD
iptables -P INPUT ACCEPT
iptables -P OUTPUT ACCEPT
iptables -P FORWARD DROP
#
# Accept ongoing connections
iptables -A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
#
# For your own safety, stop users logging in from other VMs
#
iptables -A INPUT -m conntrack --ctstate NEW -m tcp -p tcp --dport 22 ! -s
10.0.0.0/16 -j ACCEPT
```

```
iptables -A INPUT -m conntrack --ctstate NEW -m tcp -p tcp --dport 23 ! -s
10.0.0.0/16 -j ACCEPT

#
# Your changes go after here.
#
```

What is the device name for your primary network connection in your virtual machine?
$ ip route show

```
[root@host-6-73 ~]# ip route show
default via 10.0.6.78 dev ens3  proto static  metric 100
10.0.6.72/29 dev ens3  proto kernel  scope link  src 10.0.6.73  metric 100
192.168.122.0/24 dev virbr0  proto kernel  scope link  src 192.168.122.1
[root@host-6-73 ~]#
```

Device name = /dev/ens3

Add to the END of your /root/firewall script a rule which, when an http packet (tcp) comes in from your main interface, jumps to DROP. Execute the script to activate this change. Do not make this new rule stateful (so no conntrack).
$ iptables -A INPUT -p tcp --dport 80 -i ens3 -j DROP

Add another rule to the end of your /root/firewall script. This new rule jumps to DROP when a tcp packet which has s source address of 20.0.0.0/24 comes in from your main network device. Execute the script to activate this change. You will be marked wrong if your rule has more conditions than those listed in the question. Do not use connection tracking.
$ iptables -A INPUT -p tcp --dport 80 -i ens3 -s 20.0.0.0/24 -j DROP

If any packet is passed through the FORWARD chain, reject the packet with the default settings.
$ iptables -t filter -A FORWARD -j REJECT

Accept ping at a limit of 1 per second from any interface in the INPUT chain. Do not use connection tracking in the rule. DROP pings faster than the limit.

Warning: Even without conntrack specifically in the rule, you are still making connection tracking rules elsewhere in the file.
Only the first ICMP ping is NEW, and the rest are RELATED. Therefore you MUST insert your new rules BEFORE the RELATED, ESTABLISHED test, and make sure that unwanted pings never reach the RELATED test (needs 2 rules).

Double check this is working using the ping option of the firewall tests (via the “useful” tab in the control panel). You should see “limited, 1/second” if you have done this correctly.
$ iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 1/second -j ACCEPT
$ iptables -A INPUT -p icmp --icmp-type echo-request -j DROP

Continue on from the previous set of rules. Add in one more rule so that if you receive pings faster than 1 per second, those pings will be logged. Note that things getting logged will appear at the end of /var/log/messages. Do not use a new chain to do this, and keep the rule as simple as possible. Do not use connection tracking.
$ iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 2/second -j LOG

Using the firewall tests, run a ping test and then look at the information created in /var/log/messages. With this information, what is the ip source of all these ping requests?
$ cat /var/log/messages | grep ens3 | grep TYPE=8

```
[root@host-6-73 ~]# cat /var/log/messages | grep ens3 | grep TYPE=8
Nov  1 15:35:03 host-6-73 kernel: IN=ens3 OUT= MAC=00:0c:49:00:06:90:9a:15:5d:69:28:81:08:00 SRC=10.200.0.1 DST=10.0.6.73 LEN=84 TOS=0x00 PREC=0x00 TTL=63 ID=1419 DF PROTO=ICMP TYPE=8 CODE=0 ID=22593 SEQ=7
Nov  1 15:35:03 host-6-73 kernel: IN=ens3 OUT= MAC=00:0c:49:00:06:90:9a:15:5d:69:28:81:08:00 SRC=10.200.0.1 DST=10.0.6.73 LEN=84 TOS=0x00 PREC=0x00 TTL=63 ID=1562 DF PROTO=ICMP TYPE=8 CODE=0 ID=22593 SEQ=8
Nov  1 15:35:04 host-6-73 kernel: IN=ens3 OUT= MAC=00:0c:49:00:06:90:9a:15:5d:69:28:81:08:00 SRC=10.200.0.1 DST=10.0.6.73 LEN=84 TOS=0x00 PREC=0x00 TTL=63 ID=1698 DF PROTO=ICMP TYPE=8 CODE=0 ID=22593 SEQ=9
Nov  1 15:35:04 host-6-73 kernel: IN=ens3 OUT= MAC=00:0c:49:00:06:90:9a:15:5d:69:28:81:08:00 SRC=10.200.0.1 DST=10.0.6.73 LEN=84 TOS=0x00 PREC=0x00 TTL=63 ID=1730 DF PROTO=ICMP TYPE=8 CODE=0 ID=22593 SEQ=10
Nov  1 15:35:04 host-6-73 kernel: IN=ens3 OUT= MAC=00:0c:49:00:06:90:9a:15:5d:69:28:81:08:00 SRC=10.200.0.1 DST=10.0.6.73 LEN=84 TOS=0x00 PREC=0x00 TTL=63 ID=2069 DF PROTO=ICMP TYPE=8 CODE=0 ID=22593 SEQ=12
```

Question 2: Firewall: Tighter Ruleset

In this tutorial we are going to work on a strict firewall configuration of your machine. Extra-special care must be taken when doing this, or you will sudenly find you can no longer log in. Create the following as a script called firewall2 in /root.

```
#!/bin/bash
#
iptables -F INPUT
iptables -F OUTPUT
iptables -F FORWARD
iptables -P INPUT DROP
iptables -P OUTPUT DROP
iptables -P FORWARD DROP
#
#
iptables -A INPUT  -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
iptables -A OUTPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
#
# --- Put a rule here if you want to be inserting at the start of INPUT
#
# ---
#
# Make sure ssh and telnet stay working, and that users on
# other VMs cannot log in.
#
iptables -A INPUT -m conntrack --ctstate NEW -p tcp --dport ssh ! -s
10.0.0.0/16 -j ACCEPT
iptables -A INPUT -m conntrack --ctstate NEW -p tcp --dport telnet ! -s
10.0.0.0/16 -j ACCEPT
#
```

I copied the above text into nano and ran it. Everything worked fine.

Your ssh and telnet connections are proxied from 10.200.0.1. Thus when you telnet or ssh into linuxzoo, a special program catches this traffic and forwards it on your behalf. This makes ssh and telnet traffic appear to arrive at your virtual machine from a single networked address, the proxy address of 10.200.0.1.

Inster a single rule using this script, inserting at the start of the INPUT chain so that telnet is only permitted from 10.200.0.1 arriving on the device. Leave the other rules shown above unchanged.
If telenet connections arrive from anywhere else, they should be directed to REJECT. Make this a stateful rule, and check the state before any other test.
Note that if you change any of the other rules, insert more than one rule, or do anything other than insert a single rule at the start of the chain, you will always be marked wrong.

Hint: you can put the "!" character in front of a "-s" test and the rule checks that it is NOT that address.
$ iptables -A INPUT -p tcp --dport telnet ! -s 10.200.0.1 -j REJECT

Add a rule to the firewall so that if someone on your virtual machine tries to open an http connection to 10.200.0.1, the packet is ACCEPTED. Add this rule to the END of the appropriate chain. Do not change any of the existing rules.
Use a stateful rule, and check for NEW state before any other tests.
$ iptables -A OUTPUT -m conntrack --ctstate NEW -p tcp --dport http -d 10.200.0.1 -j ACCEPT

Add appropriate rules so that, if this machine was a router, it would allow RELATED and ESTABLISHED traffic to flow in both directions, as well as permitting http and ping requests to an intranet machine 192.168.1.5.
For your rules you can assume the intranet is on device eth9, and you should make sure that the NEW packets are sent on eth9.

Note: that in reality eth9 does not exist. However this should have no effect on the rules. Add only 1 rule for RELATED, ESTABLISHED testing, 1 rule for http, and 1 rule for ping (a total of 3 rules), and create them in that order. Make all rules stateful.
$ iptables -A FORWARD -m conntrack --ctstate NEW - tcp --dport http, ping
$ iptables -A FORWARD -m conntrack --ctstate NEW -p tcp --dport 80 -d 192.168.1.5 -o eth9
$ iptables -A FORWARD -m conntrack --ctstate NEW -p icmp --icmp-type 8 -d 192.168.1.5 -o eth9 -j ACCEPT

DNS named

Question 1: Basic Setup

Each DNS server needs its own particular setup when working in each particular environment. In Linuxzoo, all DNS traffic is intercepted for security reasons by the gateway server and handles via a proxy. You must update the named configuration to take this into account.

Configure the /etc/named.conf file with the new options (so put this in the options {…….} area of the file).

```
forwarders { 10.200.0.1; };
        forward only;
```

Opened up the file in nano and added the following:

```
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// See the BIND Administrator's Reference Manual (ARM) for details about the
// configuration located in /usr/share/doc/bind-{version}/Bv9ARM.html

options {
        listen-on port 53 { 127.0.0.1; };
        listen-on-v6 port 53 { ::1; };
        forwarders {10.200.0.1; };
        forward only;
        directory       "/var/named";
        dump-file       "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
```

Question 2: See it working

Start up the NAMED service (named.service) and check that it works. If you have been playing with the firewall configuration you will need to reset the firewall to the defaults. To do that use:
systemctl restart iptables

To start NAMED run “systemctl start named.service”. Note: this may take a few seconds, especially the first time, as it rebuilds its root cache.
You have to reload or restart this service when you make a configuration change remember.
The easiest way to check that the service and config files work is:
dig +timeout=3 +tries=1 localhost @localhost

If it responds then your server is up. It should indicate that the SERVER was 127.0.0.1 (or perhaps ::1 if localhost is considered an IPv6 address) and that localhost is an A record of 127.0.0.1
$ systemctl start named.service – this started almost instantly

Qusetion 3: New Zone

Create a brand new forward zone for “sillynet.net” in the named configuration directory /var/named/.

To create a new zonefile, go to /var/named and copy named.localhost to sillynet.zone (ans make sure the “named” user can read the file). In this new sillynet.zone file, remove any A or AAA lines. It should initially look something like this:

```
$TTL 1D
@       IN SOA  @ rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      @
```

The SOA and NS records are fine the way they are, and do not need to be changed. Add this file, along with its zone information, to the configuration file /etc/named.conf.

Do not delete or mangle other information in named.conf. Keep the “.” zone!

By editing the sillynet.zone file, this niew zone should provide:

Sillynet.net -> IP address 12.0.0.20

www.sillynet.net -> IP address 12.0.0.30

$ cd /var/named

$ cp named.localhost sillynet.zone

$ ls -l sillynet.zone

$ chmod o+r sillynet.zone

Below is the changes to the sillynet.zone file

```
  GNU nano 2.3.1                                          File: sillynet.zone

$TTL 1D
@        IN SOA @ rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
                        NS      @
sillynet.net.           IN A            12.0.0.20
www.sillynet.net.       IN A            12.0.0.30
```

Below is the changes to /etc/named.conf file

```
zone "sillynet.net" IN {
        type master;
        file "sillynet.zone";
        allow-update{none;};
};
```

Hints. It is a master type. The zone file must be readable by the user “named”.

Remember to use “systemctl reload named.service” when you change the config file. If you reload and it does not work, what does “systemctl -l status named.service” tell you?

Question 4: New Zone

Now you need to build a reverse zone for sillynet.zone, mapping the 12.0.0.0/24 range to these 2 new names created in the previous question. What is the zone name in in-addr.arpa format for such a zone?

Zone name = 0.0.12.in-addr.arpa

Now build a reverse zone for sillynet.zone, mapping the 12.0.0.0/24 range to these 2 new forward names created in the earlier question. Put their definitions into a file called "sillynet.rev".

The easiest way to create the initial format for sillynet.rev is to copy from "named.loopback". If you do use this file as your template, then remove the PTR and A and AAAA records from sillynet.rev before you start. Leave the other stuff as is. It should look something like this:

```
$TTL 1D
@       IN SOA  sillynet.net. rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      sillynet.net.
```

In answering this question, update this new reverse zone file to produce:

```
12.0.0.20 -> sillynet.net
12.0.0.30 -> www.sillynet.net
A nameserver definition for the zone of sillynet.net.
```

Hints: Remember to have a zone in named.conf. Dots are important. Can the named user read the new file?

Any errors in /var/log/messages?

Below is the changes to the sillynetrev.zone file

```
$TTL 1D
@       IN SOA  @ rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      sillynet.net.

20      IN      PTR     sillynet.net.
30      IN      PTR     www.sillynet.net.
```

Below is the changes to /etc/named.conf file

```
zone "0.0.12.in-addr.arpa" IN {
        type master;
        file "sillynet.rev";
        allow-update{none;};
};
```

Use dig to verify the reverse lookup is operating normally. Use “dig” with the “-x” flag, to query the reverse zone for 12.0.0.30. Make sure you send this query to your own DNS server “@localhost”.
In response to this dig, what is the IP number returned in the A resource record information returned within the “;;ADDITIONAL SECTION” glue record?
$ dig www.sillynet.com @localhost

```
[root@host-6-73 named]# dig  www.sillynet.net @localhost

; <<>> DiG 9.9.4-RedHat-9.9.4-38.el7_3.2 <<>> www.sillynet.net @localhost
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 40000
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.sillynet.net.              IN      A

;; ANSWER SECTION:
www.sillynet.net.       86400   IN      A       12.0.0.30

;; AUTHORITY SECTION:
sillynet.net.           86400   IN      NS      sillynet.net.

;; ADDITIONAL SECTION:
sillynet.net.           86400   IN      A       12.0.0.20

;; Query time: 0 msec
;; SERVER: ::1#53(::1)
;; WHEN: Fri Nov 08 15:50:05 GMT 2019
;; MSG SIZE  rcvd: 91
```

Glue record IP = 12.0.0.20

Question 5: Advanced Zone

Create a brand new forward zone for domain “advanced.com”. For this create a new forward zone file “advanced.zone” (copy named.localhost as a starting point), and a new reverse zone file “advanced.rev” (copy named.loopback as a starting point).
Add both forward and reverse zones, along with the zone information, to the configuration file /etc/named.conf.
The zone will use the 172.16.1.0/24 network, so in the named.conf file the reverse zone will be “1.16.172.in-addr.arpa”.

This zone should give:

advanced.com -> IP address 172.16.1.1
-> MX record mail.advanced.com, priority 10
-> MX record mail.offsite.com, priority 20
www.advanced.com -> IP address 172.16.1.10,
172.16.1.11,
172.16.1.12 using a round-robin selection.
mail.advanced.com -> IP address 172.16.1.1

172.16.1.1 -> advanced.com
172.16.1.10 -> www.advanced.com
172.16.1.11 -> www.advanced.com
172.16.1.12 -> www.advanced.com
A nameserver definition for the reverse zone of advanced.com.

Set the responsible person email in both SOA records to me@advanced.com.

Below is the /etc/named.conf file

```
zone "advanced.com" IN {
        type master;
        file "advanced.zone";
        allow-update{none;};
};

zone "1.16.172.in-addr.arpa" IN {
        type master;
        file "advanced.rev";
        allow-update{none;};
};
```

Below is the /var/named/advanced.zone file

```
$TTL 1D
$ORIGIN advanced.com.
@        IN SOA  @ me.advanced.com. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
         NS      @
                 IN A    172.16.1.1
         MX      10 mail.advanced.com.
         MX      20 mail.offsite.com.
www.advanced.com. IN A  172.16.1.10
www.advanced.com. IN A  172.16.1.11
www.advanced.com. IN A  172.16.1.12
mail.advanced.com. IN A 172.16.1.1
```

Below is the /var/named/advanced.rev file

```
$TTL 1D
@        IN SOA  @ me.advanced.com. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
         NS      advanced.com.

1        IN      PTR     advanced.com.
10       IN      PTR     www.advanced.com.
11       IN      PTR     www.advanced.com.
12       IN      PTR     www.advanced.com.
```

Diag

Server Admin Tasks

Question 1: Split into 2 user groups

Two users, Bob and Jim, currently share the same group id. User Jim is being moved to another project, and he needs to be given his own group id.
Modify the user account information so that Jim keeps his current user id but moves to a new group called “hoho”. You have to create that new group. Make sure his home directory and all files are changed too.
$ groups bob – shows the group bob is currently in
$ groups jim – shows the group jim is currently in
$ groupadd hoho – creates a group called hoho
$ usermod -g hoho jim – changes the user jim to the new group hoho

Question 2: User bill cannot log in

User “bill” reports that when he tries to log in, the system immediately kicks him out. He tells you that his password is “bill” (i.e. the same as his username). He accidentally left himself logged in during a lab and maybe someone did something to his account?
HINT: As this is only affecting bill, check his shell (such as .bashrc or .bash_profile) files first. You can test the fault by trying to login as bill using ssh.

Question 3: User ben cannot save his work

User “ben” reports that after he logs in, he is unable to create any files or directories in his home directory. He tells you that his password is “ben” (i.e. the same as his username).
You will need to be familiar with the chmod command used to change file permissions.
Firstly, I had to check the permissions of bens home directory to see what the current permissions were.
$ ls -ls /home/ben
I could observe that ben could read and execute files but he could not write to the directory, therefore the way to solve his issue was to give him write permissions on the directory.
$ chmod u+w /home/ben
I then tested the solution had worked by logging in as ben and creating a file within his home directory.

Question 4: User amy cannot run ls

User “amy” reports that after she logs in, the “ls” command appears to doo nothing. Her password is “amy”. Restore the ls command behaviour to normal.
HINT: Aliases can be used to change command behaviours.

Apache 1

HTTP Server

Question 1: Run the apache server

Each time you make a configuration change to the Apache server you must restart (or at the very least reload) the httpd service. Remember to start apache for the first time by using:
$ systemctl start httpd.service

And to reload the configuration file do:
$ systemctl reload httpd.start

Additionally, if you have been changing the standard firewall configuration, you should reset the config to normal. To do this do:
$ systemctl restart iptables.service
$ systemctl start httpd.service

Question 2: Add user directories

Apache allows you do have a URL which starts with /~username. This redirects the server to look for files in /home/username/public_html/. So for instance http://machine/~dave/hello.html would look for a file in /home/dave/public_html/hello.html.

To get this feature enabled you have to hunt for the configuration statements which control UserDir. Sometimes this is in /etc/httpd/conf/httpd.conf, but in Centos7 this is in /etc/httpd/conf.d/userdir.conf. Find the line:
UserDir disable

Remove this line, or better yet put a # infront of it (which comments the line out). Then look on a little further on to find:
#UserDir public_html

Delete the '#' comment character from the front of this line. As you have changed the configuration remember to reload the httpd service!

```
  GNU nano 2.3.1    File: /etc/http

#
# UserDir: The name of the directo
# directory if a ~user request is 
#
# The path to the end user account
# accessible to the webserver user
# must have permissions of 711, ~us
# of 755, and documents contained t
# Otherwise, the client will only 
#
<IfModule mod_userdir.c>
    #
    # UserDir is disabled by defaul
    # of a username on the system 
    # permissions).
    #
    #UserDir disabled

    #
    # To enable requests to /~user/
    # directory, remove the "UserDi
    # the following line instead:
    #
    UserDir public_html
</IfModule>
```

The image shows the UserDir being disabled and the UserDir public_html enabled.

$ systemctl reload httpd.service

Now create a user called "dave", create a public_html directory in dave's home directory, and create a file hello.html in the public_html. The contents of this file should be:

```
<html>
<body>
<h1>HOST</h1>
<p>
I am clever
</p>
</body>
</html>
```

$ useradd dave

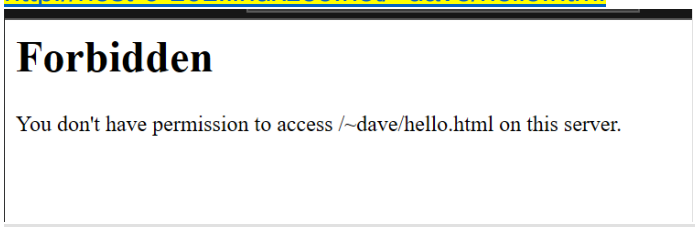
The best way to ensure that "dave" files and directories are all owned by dave is to "su - dave", make the files and directories, then CTRL-D back to root.
Otherwise do "chown -R dave.dave /home/dave". The "/home/dave" and "/home/dave/public_html" must be executable by others, and the "/home/dave/public_html/hello.html" file must be readable by other. More generous permissions or permission changes on owner or group from the default will be marked incorrect.

```
$ su – dave
$ mkdir public_html
$ cd public_html
$ nano hello.html
$ Ctrl+D
$ ls -ld /home/dave
$ ls -ld /home/dave/public_html
$ ls -l /home/dave/public_html
$ chmod o+x /home/dave
$ chmod o-r /home/dave/public_html
$ ls -ld /home/dave
```

You can direct your browser to see this page by using the URL
http://yourmachinename/~dave/hello.html

Replace "yourmachinename" with the output of running the "hostname" command, eg
[root@host-19-17 dave]# hostname
host-19-17.linuxzoo.net

http://host-6-161.linuxzoo.net/~dave/hello.html



The above image shows the output of trying to view the URL

Finally, SELinux is currently enabled in enforcing mode. This means you need even more security to overcome (configure). You need to make sure that the SELinux boolean

httpd_read_user_content is enabled. By default SELinux is forbidden from reading any file in /home. Check with:

```
getsebool  httpd_read_user_content
```

and if needed set it with

```
setsebool -P httpd_read_user_content 1
```

"setsebool" may take 20 or more seconds to run. It will finish, honest!

$ getsebool httpd_read_user_content

```
[root@host-6-161 ~]# getsebool httpd_read_user_content
httpd_read_user_content --> off
[root@host-6-161 ~]#
```

$ setsebool -P httpd_read_user_content 1

$ getsebool httpd_read_user_content

```
[root@host-6-161 ~]# getsebool httpd_read_user_content
httpd_read_user_content --> on
```

$ systemctl reload httpd.service

Question 3: Add two new directories/files

Create the following directories, each of which must be executable for other:

- /home/dave/public_html/web
- /home/dave/public_html/vm

In each of these new directories create a file called "hello.html", which are copies of hello.html from /home/dave/public_html, except in "web/hello.html" replace the word HOST with WEB. In "vm/hello.html" replace the word HOST with VM. Case is important.

$ cd /home/dave/public_html

$ cp hello.html ./vm/

$ cp hello.html ./web/

$ cd web/

$ nano hello.html

$ cd ../vm

$ nano hello.html

$ systemctl reload httpd.service

Question 4: Create 2 virtual hosts

You need to create a number of virtual hosts in your virtual machine. These should go into a new file somewhere in /etc/httpd/conf.d. For the purposes of this tutorial, create and use the file "/etc/httpd/conf.d/zvirtual.conf".

$ nano /etc/httpd/conf.d/zvirtual.conf

Using <VirtualHost> create two VirtualHosts in the file /etc/httpd/conf.d/zvirtual.conf. Below is an example of a VirtualHost definition which may help you remember what is needed.

```
<VirtualHost *:80>
   ServerAdmin me@grussell.org
```

```
   DocumentRoot /home/gordon/public_html/grussell.org
   ServerName sql.grussell.org
   ErrorLog logs/sql-error_log
   CustomLog logs/sql-access_log common
</VirtualHost>
```

The names of your virtual hosts have to be worked out by yourself from your current hostname. Type in the command "hostname" and you will get something like:
host-3-2.linuxzoo.net

our machine is known by this name in DNS. It is also known by two other names, where the word "host" has been replaced with "web" and "vm". In this example of host-3-2, this machine is also known as:

web-3-2.linuxzoo.net
vm-3-2.linuxzoo.net

IMPORTANT: Do not just copy this example, as your machine number is likely to be entirely different. Use "hostname" and work your machine names out for yourself. Note too that your hostname can change each time you reboot, so double check each time you reboot!

Once you have your web and vm machine names, create two virtual host entries, one for each of web-?-?.linuxzoo.net and vm-?-?.linuxzoo.net, so that the DocumentRoot of web is /home/dave/public_html/web and the DocumentRoot of vm is /home/dave/public_html/vm.

Each VirtualHost tagged area (you need 2) needs to be configured, with their own ServerName and DocumentRoot. like: The other fields are not important in this question.
This is the first VirtualHost area.

```
<VirtualHost *:80>

    ServerAdmin me@grussell.org

    DocumentRoot /home/dave/public_html/web

    ServerName web-6-161.linuxzoo.net

    ErrorLog logs/sql-error_log

    CustomLog logs/sql-access_log common

</VirtualHost>
```

Second VirtualHost area.

```
<VirtualHost *:80>

    ServerAdmin me@grussell.org

    DocumentRoot /home/dave/public_html/vm

    ServerName vm-6-161.linuxzoo.net

    ErrorLog logs/sql-error_log

    CustomLog logs/sql-access_log common

</VirtualHost>
```

$ systemctl reload httpd.service

It is easy to make a syntax error in the config file. If you have problems you can check for syntax errors using the command:

```
httpd -t
```

...and if you make changes to a configuration file remember to tell httpd.service!

Once again you can verify this works manually by pointing your browser to web-?-?.linuxzoo.net or the vm equivalent (remembering to put the right things in for the "?" characters). This is important, as in an assessment you may need to verify this yourself.

Question 5: Rewrite Rules

Add to the VirtualHost tag area for the "vm-?-?.linuxzoo.net" virtual host a ServerAlias for "host-?-?.linuxzoo.net". Remember to replace the ? characters with the details for YOUR machine. Once again, you can remind yourself of what this is by running the hostname command. After doing this the use of host-?-? should also use the virtual host information for vm-?-?.

Add to the end of the VirtualHost tag area for vm-?-? a rewrite rule, such that any use of host-?-?.linuxzoo.net gets an external redirection to rewrite it to vm-?-?.linuxzoo.net.

```
<VirtualHost *:80>

    ServerAdmin me@grussell.org

    DocumentRoot /home/dave/public_html/vm

    ServerName vm-6-161.linuxzoo.net

    ServerAlias host-6-161.linuxzoo.net

    RewriteEngine on

    RewriteCond %{HTTP_HOST} '^host-6-161\.linuxzoo\.net$'

    RewriteRule .* http://vm-6-161.linuxzoo.net/ [R]

    ErrorLog logs/sql-error_log

    CustomLog logs/sql-access_log common

</VirtualHost>
```

Question 6: Extend Rewrite Conditions

Modify the rewrite rules for the previous question with an additional condition, so that host-?-?.linuxzoo.net always gets rewritten using an external redirect to vm-?-?.linuxzoo.net unless the URI starts with a /~dave. Thus:

http://host-?-?.linuxzoo.net/hello.html -> rewritten to --> http://vm-?-?.linuxzoo.net/hello.html
http://host-?-?.linuxzoo.net/~dave/hello.html -> not rewritten and handled normally

```
ServerAlias host-6-161.linuxzoo.net

RewriteEngine on

RewriteCond %{HTTP_HOST} '^host-6-161\.linuxzoo\.net$'

RewriteCond %{REQUEST_URI} !^/~dave

RewriteRule .* http://vm-6-161.linuxzoo.net/ [R]

ErrorLog logs/sql-error_log
```